

TaskFlow: Fullstack Task Management System

Project Details

Role: Software Developer

Project Type: fullstack

Difficulty: junior

Duration: 14 days

Focus: api_development

Student Name: Rahul Sharma

Description: A lightweight task manager with role based API and PDF reporting.

Technologies

- JavaScript
- TypeScript
- React
- Next.js
- Node.js
- Express.js
- Laravel (PHP)
- Tailwind CSS
- MongoDB
- JWT

Requirements

- Build a practical, measurable project that matches the student's current skills and education.

Objectives

- Implement a secure REST API.
- Create a responsive Next.js UI.
- Generate PDF reports.
- Document the system.

Functional Requirements

- Register/login with email/password.
- CRUD tasks for authenticated users.
- Admin assigns tasks to any user.
- Users view only their tasks.
- Export task list as PDF.
- Return JSON with proper status codes.

Non Functional Requirements

- Hash passwords with bcrypt.
- JWT expires in 1 hour.
- Rate limit 100 req/min/IP.
- Validate all inputs, 400 on errors.
- Log errors to console.

Modules

Auth Module

Description: Handles registration, login, JWT, role assignment.

Features

- Register endpoint
- Login endpoint
- JWT middleware
- Role middleware

Task Module

Description: CRUD and assignment of tasks.

Features

- Create task
- Read tasks
- Update task
- Delete task
- Assign task

Report Module

Description: Generate PDF of task list.

Features

- PDF endpoint
- Laravel PDF service
- Download link

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Status: active